

W1 Intro DS + BIG DATA

Goal: turn raw data → info/knowledge for decisions; exam wants method justification, not tool names only.

Workflow: question → acquire → clean/transform → store → EDA → model/insight → evaluate → communicate/deploy.

Data: structured/semi/unstructured; batch/stream; small/large; observational/experimental; labelled/unlabelled.

Big data Y's: volume, velocity, variety, veracity, value; choose tech by bottleneck: storage, query, compute, governance.

Measurement: nominal=no order; ordinal=order no equal gaps; interval=equal gaps no true 0; ratio=true 0.

Pitfall: encoding categories as 1,2,3 creates fake distance/order unless ordinal is real.

Q Useful response

business decision, stakeholder, cost of errors

data unit, row meaning, leakage risk

model target, metric, validation

ethics consent, bias, harm, privacy

Problem framing: supervised pred; unsup pattern; descriptive summary; causal claim needs design/assumptions.

Data quality: accuracy, completeness, consistency, timeliness, uniqueness, validity; document assumptions.

Exam pattern: order DS workflow; identify data type; explain why a technique fits organisational context. W2 SPREADSHEET EDA

EDA: inspect shape, units, missingness, duplicates, outliers, distribution, subgroup effects.

Summary: count/mode for nominal; median/IQR for ordinal/skew; mean/SD for symmetric numeric; min/max/range for spread.

Mean sensitive to outliers/skew; median robust; mode works for nominal; IQR = Q3 - Q1.

Visual checks: bar-count categories; histogram-dist numeric; box-spread/outliers; scatter=2 numeric; line=time order.

Skew: right-skew usually mean > median > mode; left-skew mean < median < mode.

Recall focus: map mean/median/mode to skewed income/salary plots; nominal cannot be naturally ranked.

Type	Centre	Spread	Plot
nominal	mode/count	prop	bar
ordinal	median	IQR	bar/box if coded
numeric	mean/median	SD/IQR	hist/box
paired num	-	coor	scatter

Correlation: Pearson linear numeric; Spearman monotonic/rank; corr ≠ causation; outliers can dominate.

z = (x - x̄) / s standardises units; useful before distance, PCA, regularised regression.

Pitfall: high R² after cherry-picking subset/features may be leakage/overfit; validate out-of-sample. W3 PYTHON EDA

Reproducibility: code notebook/script, fixed random seed, saved env, clear data provenance, no manual spreadsheet edits.

Pandas pipeline: read → inspect dtypes/head/describe → clean names → missing/outlier handling → groupby/merge/pivot → plot.

df.isna().mean() missing rate; groupby(keys).agg() summaries; merge(..., how=) joins.

Missing: MCAR/MAR/MNAR; deletion simple but biased if not MCAR; Imputation must be fit on train only.

Encoder: one-hot nominal; ordinal map only if order meaningful; scaling numeric; text vectorisation later.

Leakage: preprocessing using whole data before split leaks test info; fit transformers on train, apply to validation/test.

Symptom	Check/fix
dtype object	parse numeric/date; strip spaces
duplicates	define key; keep latest?
impossible value	rule/range validation
class imbalance	stratified split, metric choice

Sampling: random, stratified, cluster; biased sample limits external validity.

Train/val/test: train fit params; val tune; test final estimate once.

Exam answer: always mention row unit, target definition, features available at prediction time. W4 CLEAN + STORE

ETL: extract from sources, transform schema/types/quality, load to DB/warehouse; ETL loads raw first then transforms.

Cleaning: standardise units/codes → parse dates → handle missing → de-dup → resolve conflicts → validate constraints.

Schema: entity tables w/ PK; FK enforces referential integrity; many-to-many uses junction table.

Referential integrity: FK value must match referenced PK or be NULL if allowed; prevents orphan rows.

Warehouse: fact table has measures + FKs to dimensions; star denormalised dims; snowflake normalised dims.

Star	Snowflake
dim query	wide/denorm
storage	fewer joins
use	more redundancy
simple	speed/simple BI

Recall focus: construct star schema from lecturer/student/room/date/event facts.

Date features: extract year/month/day/week; beware time zone, missing dates, leakage from future dates.

Data modelling: define grain first (one row per sale? per item? per student-course-date?) then keys/measures.

Pitfall: natural key may change; surrogate key stable but needs uniqueness rules. W5 SQL SUMMARY

SQL order: FROM JOIN → WHERE → GROUP BY → HAVING → SELECT → ORDER BY → LIMIT.

COUNT(*) counts rows incl NULL; COUNT(col) counts non-NULL; AVG/SUM ignore NULL.

Join: inner = matched rows; left = all left + NULL right; natural = join by same-named cols (dangerous if accidental).

Empty set: inner join with empty right gives 0 rows; left join with 10 left

rows gives 10 rows w/ NULL right attr.

Task	Pattern
monthly top-k	EXTRACT(MONTH FROM d) or grouped month
threshold group	ORDER BY metric DESC LIMIT k
unique attr	GROUP BY ... HAVING COUNT(*)>S
anti	SELECT DISTINCT col
	LEFT JOIN ... WHERE right.id IS NULL

Aggregation: every selected non-aggregate attr must be grouped; HAVING filters groups, WHERE filters rows.

SELECT f.title, COUNT(a.id) FROM film f JOIN film_actor fa ... GROUP BY f.id, f.title HAVING COUNT(*)>S ORDER BY f.title

Pitfall: NATURAL JOIN can silently join on unintended same names; explicit ON is safer.

Subquery: IN(empty) false for all rows; EXISTS tests rows, often clearer.

Exam SQL: state join pattern, grouping grain, aggregate, filter, ordering.

W6 HYP TEST + EVAL

H_0/H_1 : H_0 default/no effect; $p = P(\text{data as/extreme} | H_0)$; reject if $p < \alpha$.

Type I: reject true H_0 (prob α). Type II: fail reject false H_0 (prob β). Power = $1 - \beta$.

Tests: one/two-sample t for means; paired t for before/after; chi-square for categorical independence; nonparam if assumptions weak.

Recall focus: if H_0 true and rejected → Type I; at $\alpha = 0.01$, = 0.01% reject.

Metric: confusion matrix TP/FP/FN/TN; accuracy bad for imbalance; precision vs recall trade-off.

$Acc = \frac{TP+TN}{N}$, $Prec = \frac{TP}{TP+FP}$, $Rec = \frac{TP}{TP+FN}$, $F1 = \frac{2PR}{P+R}$.

Need	Metric
avoid false alarm	precision/specificity
catch positives	recall/sensitivity
balanced binary	F1/ROC
numeric pred	RMSE/MAE/R²
clustering	silhouette/SSE

CV: k-fold estimates generalisation/tunes hyperparams; detects over/underfit but does not magically fix it.

Pitfall: multiple testing inflates false positives; control via preregister, correction, validation.

Answer template: define variables/groups, H_0/H_1 , test choice + assumptions, decision rule, interpretation. W7 ASSOC + PCA

Association rules: $X \rightarrow Y$ over transactions; frequent itemsets satisfy min support; rules ranked by confidence/lift.

$supp(X \cup Y) = P(X \cup Y)$, $conf = P(Y|X) = \frac{supp(XY)}{supp(X)}$, $lift = \frac{conf}{P(Y)}$.

Apriori: frequent 1-itemsets → candidate join/prune → count support → repeat; downward closed; subset of frequent is frequent.

FP-growth: count items, sort by freq, build FP-tree, mine conditional pattern bases/trees.

Recall focus: long FP-tree/association table calculations; support/confidence interpretation for policy/action.

Dim red: PCA finds orthogonal directions of max variance; unsupervised, scale-sensitive.

Choose k by cumulative explained variance, e.g. keep PCs until $\sum_{j=1}^k \lambda_j / \sum \lambda_i \geq .95$.

PCA steps: standardise → covariance/correlation matrix → eigvec/eigval/SVD → project → validate.

Pitfall: removing 10 PCs from 100 at 95% means discarded PCs jointly explain <5%, not that 10 original features are useless.

Assoc	PCA
input	transactions
output	rules/itemsets
supervised?	no
risk	many spurious rules
	no hard interpret

W8 CLUSTERING

Clustering: unsupervised grouping; no true labels usually; evaluate compactness, separation, stability, interpretability.

k-means: choose k centroids → assign nearest → update means → repeat to convergence; minimises within-cluster SSE.

$SSE = \sum_k \sum_{x \in C_k} ||x - \mu_k||^2$; sensitive to scale, outliers, initialisation, non-spherical clusters.

Choose k: elbow in SSE, silhouette high, domain meaning; k is input, not output.

Hierarchical: start singleton/agglomerate by linkage; dendrogram cut gives clusters.

Linkage	Distance between clusters
single	min pair; chaining risk
complete	max pair; compact clusters
average	mean pair
centroid	centroid distance
Ward	min variance increase

Silhouette $s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \in [-1, 1]$; high means well matched to own cluster.

Recall focus: 1-D k-means iterations; draw dendrogram from pairwise distances; justify k from silhouette/SSE.

Preprocessing: standardise numeric variables; encode categorical carefully; remove leakage IDs.

Pitfall: clustering result is exploratory; do not claim causal/true groups without validation. W9 REGRESSION

Linear reg: $\hat{y} = \beta_0 + \sum \beta_j x_j$; coefficients are ceteris paribus effects in same units.

$SSE = \sum (y_i - \hat{y}_i)^2$, $MSE = SSE/n$, $RMSE = \sqrt{MSE}$, $R^2 = 1 - SSE/SSY$.

R²: training R² usually nondecreasing when adding predictors; adjusted R² penalises complexity.

Assumptions: linearity, independent errors, homoscedasticity, approx normal residuals for inference, no severe multicollinearity.

Diagnostics: residual plot for nonlinearity/heterosced; Q-Q for normality; VIF/correlation for collinearity; leverage/influence.

Multicollinearity: predictors correlated; individual β unstable/near 0 despite pairwise corr; predictions may still work.

Issue	Response
high variance	ridge/L2
feature selection	lasso/L1
nonlinear pattern	transform/add terms/tree
outlier	check data, robust metric
leakage	split by line/group

Simple vs multiple: simple has one predictor; multiple controls for other predictors, coefficient interpretation conditional.

Recall focus: direct SSE from given line; Spearman vs Pearson; high corr but low coefficient due collinearity.

Model answer: define target, features, split, metric, baseline, fit, diagnostics, limitation. W10 CLASSIFICATION

Classification: predict class label/probability; threshold converts score to class; calibrate if probability used for decisions.

Logistic: $p(y = 1|x) = \sigma(z) = 1/(1 + e^{-z})$, $z = \theta_0 + \theta^T x$; loss is differentiable cross-entropy.

Gradient descent: initialise θ → compute pred/loss/grad → $\theta \leftarrow \theta - \eta \nabla J$; SGD uses mini/single batches.

L2 adds $\lambda \sum \theta_j^2$ shrinkage; L1 adds $\lambda \sum |\theta_j|$ sparsity/feature selection.

Decision tree: split by highest info gain; entropy $H(Y) = -\sum p_i \log_2 p_i$; IG = $H(Y) - \sum \frac{n_i}{n} H(Y|v_i)$.

Pruning: reduce overfit by cutting weak branches; may lower train accuracy but improve generalisation.

Model	Strength	Risk
logistic	interpretable probs	linear boundary
RF	handles nonlinear/mixed	overfits
NB	stable, strong	interpretable
tree	fast/test/call	independence assumption

Naive Bayes: $P(y|x) \propto P(y) \prod_j P(x_j|y)$; use smoothing for unseen categories.

Recall focus: IG root selection, pruning reason, logistic standardisation + one GD update.

Imbalance: stratify, use precision/recall/F1, class weights/resampling, threshold tuning. W11 TEST + BAYES

Unstructured data: text/images/audio need feature extraction before standard models; prep preprocessing train-only where learned.

Text pipeline: clean/tokenise → normalise case → stopwords/stem/lemmatise → vectorise → model/evaluate.

$f(t,d,f_i,d_i) = f(t,f_i,d_i) \log(N/d_f)$; rare-in-corpus terms get higher IDF, ubiquitous terms lower IDF.

Pitfall: IDF is not a probability that term is important; frequent across all docs gets low weight.

Bag-of-words: ignores order/context; n-grams recover short phrases; sparse high-dimensional matrix common.

Spam classifier: NB/logistic/SVM; justify by labelled text, high-dimensional sparse features, precision-recall trade-off.

Text issue	Handling
raw words	min df / smoothing
negation	n-grams/features
class imbalance	PRI/F1, threshold
leakage	remove labels in headers/time
interpret	top weighted terms

Bayes classifier: prior × likelihood; categorical likelihood estimated by counts; Laplace smoothing $(n + \alpha)/(N + \alpha K)$.

For NB calculation, multiply class prior by each conditional likelihood, then normalise/compare scores.

Recall focus: animal/car categorical NB; missing category strategy: choose algorithm for spam with reasons. W12 ETHICS + COMMS

Ethics: privacy, consent, fairness, transparency, accountability, security, misuse, stakeholder harm.

Risk workflow: identify stakeholders → benefits/harms → sensitive attr/proxies → bias tests → mitigation → communicate limitations.

Fairness metrics can conflict; equal accuracy alone may hide group harms; document trade-offs and decision context.

Privacy: collect minimum necessary data, de-identify, access control, retention limits; re-identification remains possible via linkage.

Communication: match audience; report method, assumptions, metric, uncertainty, limitation, action recommendation.

Claim	Need
predictive	validation metric + baseline
causal	design/assumptions, not just corr
policy	cost/benefit + affected groups
visual	labelled axes, honest scale

Visual integrity: avoid truncated axes that mislead; show uncertainty; keep chart type aligned with data type.

Final checklist: data type? SQL grain? metric? test assumptions? leakage? overfit? interpretability? ethics?

Common finals: SQL joins/aggregates/date extract; FP/association; k-means/hierarchical; IG/tree; logistic GD; regression SSE; hypothesis test.

Best exam answers justify choices wrt data, assumptions, objective, and consequences of false decisions.

EXAM FORMULA BANK

Entropy/Gini: $H(S) = -\sum p_i \log_2 p_i$; $Gini = 1 - \sum p_i^2$; lower impurity better.

$IG(S,A) = H(S) - \sum_{v \in A} \frac{|S_v|}{|S|} H(S_v)$; root = max IG.

Regression: residual $e_i = y_i - \hat{y}_i$; $SSE = \sum (y_i - \hat{y}_i)^2$; $SSE = \sum e_i^2$; $R^2 = 1 - SSE/SSY$.

Classification: specificity = $TN/(TN + FP)$; FPR = $FP/(FP + TN)$; FNR = $FN/(FN + TP)$.

Standardisation: $x = (x - \mu)/\sigma$; train μ , σ only; apply same transform to new data.

Log odds: logistic coefficient θ_j changes log-odds by θ_j per unit, holding others fixed.

GD update: $\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_{\theta} J(\theta)$; too-large η diverges, too-small slow.

Ridge/Lasso: ridge stable w/ collinearity; lasso sparse but unstable if highly correlated predictors.

Lift: $lift(X \rightarrow Y) = P(XY)/(P(X)P(Y))$; > 1 positive assoc; ≈ 1 independent.

PCA reconstruction: keep top k components; reconstruction loss linked to discarded eigenvalues.

k-means SSE: decreases as k increases; elbow seeks diminishing returns, not a formal proof.

Silhouette: a own-cluster avg dist, b nearest-other avg dist; negative means closer to other cluster.

SQL MICRO PATTERNS

Join pattern: identify PK/FK path first; never aggregate before checking row multiplication.

SELECT key, SUM(x) FROM T WHERE row_cond GROUP BY key HAVING SUM(x)>c ORDER BY SUM(x) DESC LIMIT k

Date windows: EXTRACT(MONTH FROM d) IN (4,5,6): interval arithmetic needs DB-specific syntax.

COUNT trap: after LEFT JOIN, COUNT(*) counts preserved left rows; COUNT(right.id) counts matches.

Top-N per group: window functions if available; otherwise aggregate then rank/filter.

NULL logic: comparisons with NULL are unknown; use IS NULL, not =.

Anti-join: find rows without match; left join then where right key is NULL.

Explain query: state relation roles, join condition, grouping grain, aggregate semantics, filtering state.

MODEL SELECTION RULES

If data/task	Consider
labelled numeric target	regression; RMSE/MAE
labelled class	logistic/tree/RF/NB; F1/AUC
unlabelled numeric	k-means/PCA
transactions	RF
relational reporting	SQL/warehouse
text labels	TF-IDF + NB/logistic

Bias-variance: underfit = high bias, train+val poor; overfit = high variance, train good val poor.

Fix underfit: richer features/models, more regularisation, train longer; fix overfit: more data, regularise, prune, simpler model, CV.

Baseline: majority class, mean predictor, simple SQL summary; model must beat useful baseline.

Metric choice: align metric with cost; do not optimise accuracy when minority positivity is important.

Cross-validation estimates performance/tunes hyperparams; final test set remains untouched until the end.

Interpretability: coefficients, rules, feature importance, partial dependence, always explain limitations.

HIGH-YIELD PITFALLS

Correlation: Pearson high but regression coeff near 0 may be multicollinearity/suppression; inspect VIF/corr matrix.

PCA: PCs are linear combos, not original variables; signs arbitrary; supervised labels not used.

Association: support/confidence denominators differ; rule $A \rightarrow B$ and $B \rightarrow A$ share support but not confidence.

Hyper test: fail to reject H_0 is not proof H_0 true; report uncertainty/effect size.

Decision tree: max IG greedily local; tree can overfit noisy attributes; pruning trades train fit for generalisation.

Logistic: output is probability, not continuous target; threshold should reflect cost/imbalance.

NB: multiply many small probabilities; in implementation use log probabilities, smoothing avoids zeroing class score.

Clustering: IDs and unscaled dollar variables can dominate Euclidean distance; validate clusters with domain meaning.

SQL: WHERE cannot use aggregate aliases in standard SQL; use HAVING or subquery.

Ethics: de-identified data can be re-identified; proxies can encode sensitive attributes.

ANSWER TEMPLATES

Explain method: define objective → assumptions → steps → metric/output → limitation.

Calculation answer: write formula, plug numbers, compute, interpret in context and units.

Compare methods: data type, supervision, assumptions, computational cost, interpretability, failure mode.

SQL answer: first give query, then one sentence explaining joins, groups, and filters.

Hypothesis answer: variables/groups, H_0/H_1 , test, assumptions, α , decision, real-world wording.

Ethics answer: stakeholder, harm, evidence of risk, mitigation, residual limitation.

For COMP5310 finals, broad application beats memorised definitions: justify why this method fits this data and decision.

MINI WORKED PATTERNS

Confusion matrix: predicted rows/cols vary by question; label TP carefully before formulas.

NB score: for class c, score = $P(c) \prod_j P(x_j|c)$; choose max score; normalise only if probability requested.

One GD step, with $\theta = 0$, logistic $p = 0.5$ for all rows; gradient sign shows which weights increase.

K-means ID: assign by nearest centroid midpoints; centroid update = arithmetic mean of assigned points.

FP-tree: sort each aggregation by global frequent-item order; share prefixes; conditional base from paths ending in item.

Star schema: fact grain first; dim tables second; measures are additive/semi-additive/non-additive.

Skew summary: if outliers pull mean away, use median/IQR and show boxplot/hist.

Model report: include baseline, validation split, metric, key errors, and deployment constraints.

CALCULATION SHORTCUTS

Confusion calc: first draw 2×2 table; mark actual positive/negative and predicted positive/negative before computing metrics.

Entropy calc: for binary $p, 1-p$.

$H = -p \log_2 p - (1-p) \log_2 (1-p)$; pure node $H = 0$.

IG calc: compute parent entropy once; for each feature, weighted child entropy; highest reduction wins.

K-means calc: distance table → assignments → new means; stop when assignments/centroids unchanged.

Hierarchical calc: after each merge, update distances using chosen linkage; dendrogram height = merge distance.

Association calc: numerator for support/conf is count containing both antecedent and consequent.

PCA calc: cumulative ratio of variance explained by principal variables use different scale.

SSE calc: never forget square!

Logistic calc: compute z then sigmoid; threshold can move.

PAST-PAPER EMPHASIS

Very frequent: SQL joins, clustering, decision tree IG.

Frequent: hypothesis error by group, interpretation; data types.

Conceptual: star vs snowflake; DB-specific syntax.

L2: Pearson vs Spearman.

Application: choose model for spam/test, policy from association rules, evaluate model under stakeholder costs.

2025-style breadth: expect mixed MCQ/short/calc within same question; revise by topic links, not only by section type.

When stuck, write the definition + denominator + decision criterion; many marks come from correct setup.

ROBUSTNESS CHECKS

Data split: random split for fit; temporal split for forecasting/deployment; group split if same entity appears repeatedly.

Imbalance: report class distribution; use stratified folds; inspect confusion matrix not just accuracy.

Outliers: decide data error vs genuine rare case; robust summaries may be more defensible than deletion.

Feature selection: avoid peeking at validation/test; feature search must be inside CV loop.

Hyperparams: k in k-means, tree depth, regularisation λ , learning rate η ; tune via validation.

Generalisation: similar train/test scores good; large gap suggests overfit; both poor suggests underfit/noisy task.

Uncertainty: use CI/SE where appropriate; single metric without uncertainty can overstate evidence.

ONE-LINE DEFINITIONS

Veracity: trustworthiness/noise/bias of data. Validity: values satisfy rules/meaning.

Cardinality: number of distinct values; high-card categorical needs careful encoding.

Granularity: level of detail in rows; mismatched grain causes double counting.

Sparsity: many zeros/missing combos; common in text/transactions.

Curse of dimensionality: distances less informative as dimensions grow, PCA/feature selection may help.

Calibration: predicted probabilities match empirical frequencies.

Threshold: decision boundary on score; chosen by cost, not fixed by model.

Proxy variable: non-sensitive feature correlated with sensitive attribute; can carry bias.

Audit trail: records data/source/model decisions; supports reproducibility/accountability.

FINAL CHECKLIST

Before SQL: identify row grain and whether joins duplicate rows; decide if filtering rows or groups.

Before model: identify target availability time; remove future/ID/leaky data; choose baseline and metric.

Before test: check independent vs paired samples; numeric vs categorical outcome; one/two-sided question.

Before PCA/clustering: scale variables, remove identifiers, decide whether interpretability loss is acceptable.

Before ethics answer: name affected group, concrete harm, mitigation, and remaining limitation.

Common denominators: support uses all transactions; confidence uses antecedent count; recall uses actual positives; precision uses predicted positives.

Common outputs: regression gives numeric $\hat{y}$; logistic gives probability; tree gives rules/probs; clustering gives group labels; PCA gives components.

Ambiguous wording: if question says "best", justify with data type + objective + assumptions; if says "explain", define and apply to scenario.

Show working for calculations: formula + substitution + final value + interpretation. Hidden arithmetic with no setup loses marks.

Denominator sanity: $P(A|B)$ denominator is count/prob of B; error rates condition on actual state; precision conditions on predictions.

Unit sanity: RMSE in target units; R²/corr unitless; support/confidence proportions: number in bits if log base 2.

Direction sanity: high recall lowers FN; high precision lowers FP; high regularisation shrinks coefficients; high k lowers k-means SSE.

Last-minute triage: secure SQL + association + tree/clustering formulas first, then polish conceptual ethics/model-choice wording.